



Linux for DevOps

Complete Study Guide

Sessions 1 through 8 + Extra Commands

Navigation · Files · System Monitoring · Processes · Networking

Text Processing · User Management · File Permissions

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Session 1

Navigation & File System Basics

pwd · cd · ls · mkdir · touch · Hidden Files

1.1 pwd — Print Working Directory

```
$ pwd  
# Output: /home/ubuntu
```

1.2 cd — Change Directory

```
$ cd linux # relative path – from current location  
$ cd .. # go back one level  
$ cd ../../ # go up two levels  
$ cd ~ # go to home directory  
$ cd - # go to previous directory (undo navigation)  
$ cd /home/ubuntu/ # absolute path – from root /
```

Type	Example	Meaning
Relative	cd linux/	From current location
Absolute	cd /home/ubuntu	Full path from root
Home	cd ~	Always /home/ubuntu
Back	cd -	Undo last cd

1.3 ls — List Files and Folders

```
$ ls # simple list  
$ ls -l # long format – permissions, size, date  
$ ls -a # show hidden files (. files)  
$ ls -la # hidden + long format  
$ ls -lt # sort newest first  
$ ls -ltr # sort oldest first (reverse)  
$ ls -R # recursive – show all subfolders  
$ ls -lh # human readable file sizes (KB, MB)
```

Flag	Meaning
-l	Long format: permissions, links, owner, size, date

Flag	Meaning
-a	All files including hidden (starting with .)
-h	Human readable sizes
-t	Sort by time, newest first
-r	Reverse sort order
-R	Recursive — list all subdirectories

1.4 mkdir & touch — Create Files and Folders

```
$ touch devops.txt # create empty file
$ mkdir linux # create single folder
$ mkdir linux unix amazing-devops # create multiple folders
$ mkdir -p session-1/commands # create nested folders (-p = parents)
$ mkdir -p Devops/{linux,git,docker} # create multiple subfolders at once

✗ mkdir -p Devops{linux,git,docker} # WRONG — missing /
✓ mkdir -p Devops/{linux,git,docker}
```

1.5 Hidden Files

Files starting with . (dot) are hidden. Use ls -a to see them.

Hidden File	Purpose
.bashrc	Terminal config — auto-runs on every new terminal
.bash_history	All commands you've ever typed
.profile	Login settings
.ssh/	SSH keys and known hosts
.	Current directory shortcut
..	Parent directory shortcut

Session 2

File Operations & Package Management

vi · cp · mv · rm · APT · Docker · SSH

2.1 vi Editor

```
$ vi filename.txt # open or create file
$ cat filename.txt # read file without opening editor
$ > filename.txt # empty/clear a file completely
```

Key	Action
i	Enter Insert mode — start typing
Esc	Exit Insert mode → go to Command mode
:w	Save the file
:q	Quit vi
:wq	Save and quit
:q!	Quit WITHOUT saving (force quit)
:set nu	Show line numbers
dd	Delete current line
yy	Copy current line
p	Paste copied line
/word	Search for 'word' in file
n	Jump to next search result

2.2 cp — Copy Files and Folders

```
$ cp devops.txt session-1/ # copy file to folder
$ cp devops.txt session-1/copy.txt # copy and rename
$ cp -r session-1/ unix/ # copy folder recursively

✗ cp devops.txt # WRONG — missing destination
✗ cp session-1/ unix/ # WRONG — folder needs -r flag
✓ cp -r session-1/ unix/
```

2.3 mv — Move and Rename

```
$ mv file.txt unix/ # MOVE — file goes into unix/ folder
```

```
$ mv oldname.txt newname.txt # RENAME — same folder, new name
```

2.4 rm — Delete Files and Folders

WARNING: Linux has NO recycle bin. Deleted files are GONE permanently!

```
$ rm file.txt # delete a file

$ rm -r folder/ # delete folder and all contents

$ rm -fr folder/ # force delete (no confirmation)

$ rm -i file.txt # interactive — asks before deleting

✗ rm session-1/ # WRONG — needs -r for folders

✓ rm -r session-1/
```

2.5 APT Package Manager

```
$ sudo apt update # update package list from internet

$ sudo apt install docker.io -y # install package (-y = no prompt)

$ sudo apt install pkg1 pkg2 -y # install multiple packages

$ sudo apt remove packagename # remove a package

$ sudo apt upgrade # upgrade all installed packages

$ apt list --upgradable # show available upgrades

✗ apt install tree # WRONG — need sudo

✓ sudo apt install tree
```

2.6 SSH — Connect to EC2

```
PS> ssh -i 'key.pem' ubuntu@ec2-xx-xx.compute-1.amazonaws.com

# First time: type 'yes' to trust the host

# $ = normal user prompt

# # = root user prompt
```


Session 3

System Monitoring & Environment

Redirection · env · wc · Logs · tail

3.1 Redirection Operators

```
$ echo 'hello' > file.txt # overwrite file with output
$ ls -l >> file.txt # append output to file
$ ls-l 2>> file.txt # append errors to file
$ ls -l >> file.txt 2>&1 # stdout + stderr into same file
```

Operator	What it does
>	Overwrite — replaces entire file
>>	Append — adds to end of file
2>	Write errors to file (overwrite)
2>>	Append errors to file
2>&1	Send stderr to same place as stdout

3.2 Environment Variables

```
$ env # show all environment variables
$ printenv # same as env

$ export SESSION=devops # set a new variable
$ echo $SESSION # print variable value
$ echo session # prints literal text 'session'

$ echo $PATH # print PATH variable
$ source ~/.bashrc # reload .bashrc without restarting

# Make variables permanent — add to ~/.bashrc:
export SESSION=devops
alias pwf='pwd | rev | cut -d "/" -f 1 | rev'
```

Variable	Value / Meaning
\$PATH	Directories Linux searches for commands
\$USER	Current logged-in username
\$HOME	Home directory (/home/ubuntu)
\$PWD	Current working directory

Variable	Value / Meaning
\$SHELL	Current shell (/bin/bash)
\$LOGNAME	Login name

3.3 wc — Word Count | who — Logged Users | Pipe |

```
$ wc /etc/passwd # lines, words, chars: 36 52 1910

$ wc -l /etc/passwd # lines only: 36

$ wc -w /etc/passwd # words only: 52

$ wc -c /etc/passwd # characters: 1910

$ who # show logged-in users

$ whoami # show current username

$ who | wc # pipe: who output → wc counts it

$ ls | wc -l # count number of files

$ ps -eaf | grep nginx # find nginx process
```

3.4 Log Files & tail

```
$ cd /var/log/ # go to log directory

$ cat /var/log/syslog # read system log

$ tail file.txt # show last 10 lines

$ tail -n 20 file.txt # show last 20 lines

$ tail -f /var/log/syslog # LIVE follow – updates in real time

$ head file.txt # show first 10 lines

$ head -30 file.txt # show first 30 lines

$ head -30 file | tail # show lines 21-30 (middle section)
```

Log File	What it records
/var/log/syslog	General system messages
/var/log/auth.log	Login and authentication events
/var/log/kern.log	Kernel messages
/var/log/apt/history.log	APT install/upgrade history
/var/log/dpkg.log	Package install/remove events

Session 4

Process Management & Jenkins

history · ps · nohup · kill · wget · Java

4.1 history — View Past Commands

```
$ history # show all past commands with numbers
$ history | tail -20 # show last 20 commands
$ history | grep wget # search history for specific command
$ !264 # re-run command number 264
$ !! # re-run last command
```

4.2 ps — Process Status

```
$ ps -eaf # show ALL running processes
$ ps -eaf | grep nginx # find nginx process
$ ps -eaf | grep java # find Java/Jenkins process
```

Column	Meaning
UID	User who owns the process
PID	Process ID — unique number
PPID	Parent Process ID
STIME	Start time
TTY	Terminal it's running on
CMD	The actual command running

4.3 nohup & Background Jobs

```
$ nohup java -jar jenkins.war &
# nohup = keep running even if SSH disconnects
# & = run in background, gives terminal back
# Output saved to nohup.out automatically

$ cat nohup.out # read background process output
$ tail -f nohup.out # live monitor background process
$ jobs # list background jobs
$ fg # bring background job to foreground
```

```
$ bg # send job to background
```

4.4 kill — Stop a Process

```
$ kill -9 11830 # force kill process with PID 11830

$ kill -15 11830 # graceful shutdown (default)

$ kill -l # list all kill signals

$ pkill nginx # kill process by name

$ killall nginx # kill all processes named nginx
```

Signal	Number	Meaning
SIGKILL	-9	Force kill — cannot be ignored
SIGTERM	-15	Graceful shutdown — asks process to stop
SIGHUP	-1	Reload config without restart

4.5 wget — Download Files

```
$ wget https://example.com/file.zip # download file

$ wget -O newname.zip https://example.com # download with custom name

$ time wget https://example.com/file # measure download time

$ wget -c https://example.com/file # resume interrupted download
```

4.6 Java Version & Jenkins

```
$ java --version # check Java version

$ sudo apt install -y openjdk-17-jdk # install Java 17

$ java -jar jenkins.war # run Jenkins (foreground)

$ nohup java -jar jenkins.war & # run Jenkins (background)

# Access: http://:8080

# Password: cat /home/ubuntu/.jenkins/secrets/initialAdminPassword
```

Session 5

Networking & File Search

find · ping · curl · tar · systemctl · HTTP Codes

5.1 find — Search Files and Folders

```
$ find . -empty # find all empty files/folders

$ find . -type f # find files only

$ find . -type d # find directories only

$ find . -name devops.txt # find by exact filename

$ find . -name '*.txt' # find all .txt files

$ find . -name '*.txt' | wc -l # count .txt files

$ find . -name sed -type f # find file named 'sed'

$ find . -empty -exec rm -rf {} \; # delete all empty files

$ find . -type f -exec grep -l 'unix' {} \; # find files containing 'unix'
```

5.2 ping — Test Network Connectivity

```
$ ping google.com # test if google.com is reachable

$ ping -c 5 google.com # send only 5 packets then stop

$ ping 8.8.8.8 # ping by IP address
```

Output	Meaning
time=1.81 ms	Round trip time — lower is better
ttl=106	Time To Live — hops remaining
0% packet loss	All packets reached the server
icmp_seq=1	Sequence number of each packet

5.3 curl — HTTP Requests

```
$ curl https://google.com # fetch URL content

$ curl localhost # test local web server

$ curl -w '%{http_code}' localhost # show HTTP status code

$ curl --connect-timeout 3 https://example.com # timeout after 3 seconds

$ curl -L https://example.com # follow redirects

$ curl -o file.zip https://example.com/file # save output to file

$ curl -I https://google.com # show headers only
```

```
✗ curl https//:google.com # WRONG — colon+slashes wrong order
✓ curl https://google.com
```

Code	Name	Meaning
200	OK	Success — page loaded
201	Created	Resource created (POST)
301	Moved Permanently	Permanent redirect
302	Found	Temporary redirect
400	Bad Request	Your request has an error
401	Unauthorized	Login required
403	Forbidden	No permission
404	Not Found	Page doesn't exist
500	Internal Server Error	Server crashed
502	Bad Gateway	Server got bad upstream response
503	Service Unavailable	Server down/overloaded

5.4 tar — Archive and Compress

```
$ tar -czvf archive.tar.gz folder/ # create compressed archive
$ tar -xvf archive.tar.gz # extract archive
$ tar -tzvf archive.tar.gz # list contents without extracting
$ tar -cjvf archive.tar.bz2 folder/ # compress with bzip2
```

Flag	Meaning
-c	Create new archive
-x	Extract archive
-z	Use gzip compression (.gz)
-j	Use bzip2 compression (.bz2)
-v	Verbose — show files being processed
-f	Specify filename (always last before filename)
-t	List contents of archive

5.5 systemctl — Manage Services

```
$ systemctl status nginx # check if nginx is running
$ sudo systemctl start nginx # start nginx
$ sudo systemctl stop nginx # stop nginx
$ sudo systemctl restart nginx # restart nginx
$ sudo systemctl enable nginx # auto-start on boot
```

```
$ sudo systemctl disable nginx # don't auto-start on boot  
$ systemctl list-units --type=service # list all services
```

Session 6

Text Processing — grep & sed

grep · regex · sed · tr · cut · rev

6.1 grep — Search Text in Files

```
$ grep 'unix' grep.txt # find lines containing 'unix'

$ grep -c 'unix' grep.txt # count matching lines

$ grep -n 'unix' grep.txt # show line numbers

$ grep -l 'unix' grep.txt # show filename only

$ grep -v 'unix' grep.txt # invert — lines WITHOUT unix

$ grep -i 'unix' grep.txt # case insensitive

$ grep -r 'unix' folder/ # search recursively

$ grep 'ubuntu' /etc/passwd # search system file

$ ps -eaf | grep nginx # find process by name

$ cat file.txt | grep 'error' # grep from pipe
```

6.2 grep Regular Expressions (Regex)

```
$ grep '^unix' file.txt # lines STARTING with 'unix'

$ grep 'unix$' file.txt # lines ENDING with 'unix'

$ grep '^unix$' file.txt # lines that are EXACTLY 'unix'

$ grep '^$' file.txt # empty lines only

$ grep -v '^$' file.txt # remove empty lines

$ grep '.' file.txt # lines with at least 1 character
```

Regex	Meaning
^	Start of line
\$	End of line
^word	Line starts with 'word'
word\$	Line ends with 'word'
^word\$	Line is exactly 'word'
^\$	Empty line
.	Any single character
.*	Any number of any characters

6.3 sed — Stream Editor

```
$ sed 's/unix/linux/' file.txt # replace FIRST match per line
$ sed 's/unix/linux/g' file.txt # replace ALL matches (global)
$ sed -i 's/unix/linux/g' file.txt # -i = modify original file
$ sed '1,3 s/unix/linux/g' file.txt # replace only on lines 1-3
$ sed '3,$ s/unix/linux/g' file.txt # from line 3 to end
$ sed -n '3p' file.txt # print only line 3
$ sed -n '$p' file.txt # print only last line
$ sed '/used/d' file.txt # delete lines containing 'used'
$ sed '4,$d' file.txt # delete from line 4 to end

✗ sed 'used/d' file.txt # WRONG — missing opening /
✓ sed '/used/d' file.txt
```

6.4 tr — Translate Characters | cut — Extract Columns | rev — Reverse

```
$ tr [a-z] [A-Z] < file # convert to UPPERCASE
$ tr [A-Z] [a-z] < file # convert to lowercase
$ tr -s ' ' < file # squeeze multiple spaces to one
$ tr -d 'x' < file # delete character 'x'

$ cut -d ':' -f 1,3 /etc/passwd # extract columns 1 & 3 (: delimiter)
$ pwd | cut -d '/' -f 5 # extract 5th path segment

$ pwd | rev # reverse the text
$ pwd | rev | cut -d '/' -f 1 | rev # get current folder name
```

Session 7

awk & User Management

awk · tee · useradd · passwd · groupadd · usermod

7.1 tee — Write to File AND Show on Screen

```
$ ls -l | tee test.txt # show on screen + save to file
$ ls -l | tee -a test.txt # show on screen + APPEND to file
$ du -sh | tee disk.txt # save du output + show on screen
```

Command	Screen	File	Mode
cmd > file	No	Yes	Overwrite
cmd >> file	No	Yes	Append
cmd tee file	Yes	Yes	Overwrite
cmd tee -a file	Yes	Yes	Append

7.2 awk — Data Processing Tool

awk processes text line by line, column by column. Syntax: awk 'CONDITION {ACTION}' file

```
$ awk '/manager/ {print}' employee.txt # print lines with 'manager'
$ awk '/manager/ {print $1}' employee.txt # print column 1 of matches
$ awk '/manager/ {print $1,$3}' employee.txt # print cols 1 and 3
$ awk '{print $NF}' employee.txt # print last column
$ awk '$NF > 25000 {print}' employee.txt # filter by value
$ awk '{print NR,$1}' employee.txt # print with line numbers
$ awk 'NR==3,NR==6 {print}' employee.txt # print lines 3 to 6
$ awk 'NR>=3 && NR<=6 && $NF>25000 {print}' employee.txt
$ awk -F: '{print $1,$3}' /etc/passwd # colon delimiter
$ awk -F: 'BEGIN {print "Name ID"} {print $1,$3}' /etc/passwd
$ awk -F: '$3 > 500 {printf "%s %s\n",$1,$3}' /etc/passwd
```

Variable	Meaning
\$1,\$2,\$3...	Column number
\$NF	Last column (Number of Fields)
NR	Current line/record number
NF	Number of fields in current line
FS	Field separator (default = space)

7.3 User Management

```
$ sudo su # switch to root user
$ sudo useradd srujan # create new user
$ sudo passwd srujan # set user password
$ su srujan # switch to user srujan
$ sudo userdel srujan # delete user
$ sudo userdel -r srujan # delete user + home directory
$ cat /etc/passwd # view all users
$ id srujan # show user ID and group info
```

7.4 Group Management

```
$ sudo groupadd devops # create group
$ sudo usermod -a -G devops srujan # add user to group
$ cat /etc/group # view all groups
$ cat /etc/group | grep srujan # check group membership
$ groups srujan # show user's groups
```

Session 8

File Permissions & Ownership

chmod symbolic · chmod numeric · chown

8.1 Understanding Permissions

```
$ ls -l
-rwxr-xr-- 1 ubuntu ubuntu 6490 abc.txt

|||■■■■■■■■■
||| u g o
||■■■■■
|| user
|■■ type: - = file, d = directory
```

Symbol	Name	Meaning
r	Read	Can view file content
w	Write	Can modify file
x	Execute	Can run as program
-	None	That permission is denied
u	User	Owner of the file
g	Group	Group that owns the file
o	Others	Everyone else
a	All	u + g + o combined

8.2 chmod — Symbolic Mode

```
$ chmod o+x abc.txt # add execute for others
$ chmod ug-w abc.txt # remove write from user+group
$ chmod ugo+rw abc.txt # add read+write for all
$ chmod a+x script.sh # add execute for all (a = ugo)
$ chmod ugo+w,ugo-x abc.txt # multiple changes at once

✗ chmod x+ug abc.txt # WRONG — who must come first
✓ chmod ug+x abc.txt
```

8.3 chmod — Numeric Mode

r=4, w=2, x=1, -=0. Add values together for each user type.

Value	Permissions	Calculation
7	rwX	4+2+1
6	rw-	4+2+0
5	r-X	4+0+1
4	r--	4+0+0
3	-wX	0+2+1
2	-w-	0+2+0
1	--X	0+0+1
0	---	0+0+0

```
$ chmod 777 file # rwxrwxrwx – full access (dangerous!)
$ chmod 755 file # rwxr-xr-x – standard for scripts
$ chmod 644 file # rw-r--r-- – standard for files
$ chmod 600 file # rw----- – private files
$ chmod 444 file # r--r--r-- – read only
$ chmod 400 file # r----- – read-only private (PEM keys)
$ chmod 741 file # rwxr----x
$ chmod -R 777 folder/ # apply recursively to folder

✗ chmod 424242 file # WRONG – only 3 digits
✓ chmod 777 file
```

8.4 chown — Change Ownership

```
$ sudo chown root:root abc.txt # change to root
$ sudo chown ubuntu:ubuntu file.txt # change to ubuntu
$ sudo chown srujan:srujan file.txt # change to srujan
$ sudo chown -R srujan:srujan folder/ # recursive ownership
```

Extra — Must-Know Linux Commands

Network · System Info · Disk · Cron · vim · Bonus Tools

E.1 System Information Commands

```
$ uptime # server uptime and load average
$ uname -a # kernel and architecture info
$ cat /etc/os-release # OS name and version
$ hostname # server hostname
$ hostname -I # server IP addresses
$ lscpu # CPU info – cores, architecture
$ lsmem # memory info
$ arch # architecture (x86_64, arm64)
$ date # current date and time
$ cal # calendar for current month
$ timedatectl # time zone and clock info
```

E.2 Disk & Memory Monitoring

```
$ df -h # disk space on all partitions
$ df -h / # disk space for root partition only
$ free -h # RAM and swap memory usage
$ du -sh folder/ # total size of a folder
$ du -h * # size of each item in current folder
$ du -h | sort -rh | head # top 10 largest folders
$ lsblk # list all disk partitions
```

E.3 Process Monitoring

```
$ top # live process monitor (q to quit)
$ htop # better interactive process viewer
$ ps -eaf # snapshot of all processes
$ ps aux # processes with memory/cpu %
$ ps aux | sort -k3 -rn | head # top CPU-consuming processes
$ pgrep nginx # find PID of nginx
```

```
$ pstree # show process tree

$ vmstat # virtual memory statistics

$ iostat # CPU and disk I/O statistics
```

E.4 Network Commands

```
$ ping google.com # test connectivity

$ ping -c 4 google.com # send only 4 packets

$ curl https://google.com # HTTP request

$ curl -I https://google.com # show HTTP headers only

$ wget https://example.com/file # download file

$ netstat -tulnp # show open ports and services

$ ss -tulnp # modern replacement for netstat

$ ifconfig # network interface info

$ ip addr # modern: show IP addresses

$ ip route # show routing table

$ traceroute google.com # show network hops to destination

$ nslookup google.com # DNS lookup

$ dig google.com # detailed DNS info

$ ssh-keygen -t rsa # generate SSH key pair

$ scp file.txt user@host:/path # secure copy to remote server

$ rsync -avz folder/ user@host:/path # sync files to remote
```

E.5 File Compression & Archives

```
$ tar -czvf archive.tar.gz folder/ # create gzip archive

$ tar -xvf archive.tar.gz # extract archive

$ tar -tzvf archive.tar.gz # list archive contents

$ zip -r archive.zip folder/ # create zip file

$ unzip archive.zip # extract zip file

$ gzip file.txt # compress single file

$ gunzip file.txt.gz # decompress gzip file
```

E.6 Text Processing Extras

```
$ sort file.txt # sort lines alphabetically

$ sort -r file.txt # sort in reverse

$ sort -n file.txt # sort numerically
```

```
$ sort -k2 file.txt # sort by column 2

$ uniq file.txt # remove consecutive duplicate lines

$ sort file.txt | uniq # remove all duplicates

$ uniq -c file.txt # count occurrences of each line

$ diff file1.txt file2.txt # compare two files

$ comm file1.txt file2.txt # compare sorted files

$ wc -l file.txt # count lines

$ nl file.txt # number lines

$ column -t file.txt # format as aligned table

$ paste file1.txt file2.txt # merge files side by side

$ xargs rm < filelist.txt # run command on each line of file
```

E.7 Cron — Schedule Automated Tasks

cron runs commands automatically at scheduled times.

```
$ crontab -e # edit your cron jobs

$ crontab -l # list current cron jobs

$ crontab -r # remove all cron jobs

# Cron syntax:

# MIN HOUR DAY MONTH WEEKDAY COMMAND

# * * * * * command

0 2 * * * /home/ubuntu/backup.sh # run at 2:00 AM daily

*/5 * * * * df -h >> /tmp/disk.log # every 5 minutes

0 0 * * 0 find /tmp -mtime +7 -delete # weekly cleanup
```

Field	Range	Meaning
MIN	0-59	Minute
HOUR	0-23	Hour
DAY	1-31	Day of month
MONTH	1-12	Month
WEEKDAY	0-7	Day of week (0=Sunday)
*	any	Run every unit
*/5	any	Run every 5 units

E.8 vim Shortcuts (Advanced vi)

Shortcut	Action
i	Insert mode — start typing

Shortcut	Action
Esc	Back to command mode
:wq	Save and quit
:q!	Quit without saving
dd	Delete current line
5dd	Delete 5 lines
yy	Copy (yank) current line
p	Paste after cursor
u	Undo last change
Ctrl+r	Redo
/word	Search for 'word'
n	Next search result
N	Previous search result
:%s/old/new/g	Replace all in file
gg	Go to first line
G	Go to last line
:20	Go to line 20
:set nu	Show line numbers
:set nonu	Hide line numbers

E.9 alias — Create Custom Shortcuts

```
$ alias ll='ls -la' # shortcut for ls -la

$ alias cls='clear' # Windows-style clear

$ alias gs='git status' # git shortcut

$ alias pwf='pwd | rev | cut -d "/" -f 1 | rev' # show folder name only

$ alias myip='curl ifconfig.me' # show public IP

# Add to ~/.bashrc to make permanent:

$ vi ~/.bashrc

# Paste aliases at bottom, then:

$ source ~/.bashrc

# Alias syntax rules:

✗ alias ll = 'ls -la' # WRONG — spaces around =

✓ alias ll='ls -la' # CORRECT — no spaces, quotes required
```

E.10 Useful One-Liners for DevOps

```
# Find top 10 largest files:

$ find / -type f -printf '%s %p\n' 2>/dev/null | sort -rn | head

# Count files in a directory:

$ ls -1 | wc -l

# Watch a command every 2 seconds:

$ watch -n 2 df -h

# Show last 100 lines of a log and follow:

$ tail -100f /var/log/syslog

# Find all files modified in last 24 hours:

$ find . -mtime -1 -type f

# Check if a port is open:

$ nc -zv google.com 80

# Show disk usage sorted largest first:

$ du -sh * | sort -rh | head -20

# Replace text in multiple files:

$ find . -name '*.txt' -exec sed -i 's/old/new/g' {} \;

# Count occurrences of a word in file:

$ grep -o 'word' file.txt | wc -l

# Run last command as sudo:

$ sudo !!

# Create file with content in one line:

$ echo 'Hello World' > newfile.txt
```

Master Cheat Sheet — All Commands

Command	Category	What it does
pwd	Navigation	Show current directory
cd folder/	Navigation	Enter folder
cd ..	Navigation	Go back one level
cd ~	Navigation	Go to home directory
cd -	Navigation	Go to previous directory
ls -l	Listing	Detailed file list
ls -a	Listing	Show hidden files
ls -lh	Listing	Human readable sizes
ls -ltr	Listing	Sort oldest first
ls -R	Listing	Recursive list
touch file	Files	Create empty file
mkdir folder	Files	Create folder
mkdir -p a/b/c	Files	Create nested folders
vi filename	Files	Open editor
cat filename	Files	Read file content
> filename	Files	Empty a file
cp file dest/	Copy/Move	Copy file
cp -r dir/ dest/	Copy/Move	Copy folder
mv file dest/	Copy/Move	Move file
mv old new	Copy/Move	Rename file
rm file	Delete	Delete file (permanent)
rm -r folder/	Delete	Delete folder
rm -fr folder/	Delete	Force delete folder
export VAR=val	Environment	Set variable
echo \$VAR	Environment	Print variable value
env	Environment	Show all variables
source ~/.bashrc	Environment	Reload config
alias name='cmd'	Shortcuts	Create command shortcut
cmd > file	Redirection	Save output to file
cmd >> file	Redirection	Append output to file
cmd 2>> file	Redirection	Append errors to file
cmd1 cmd2	Pipe	Chain commands
cmd tee file	Tee	Show + save to file

Command	Category	What it does
wc -l file	Analysis	Count lines
who	System	Show logged in users
whoami	System	Show current username
uptime	System	Server uptime
uname -a	System	Kernel info
df -h	Disk	Disk usage
free -h	Memory	RAM usage
du -sh	Disk	Directory total size
top	Monitor	Live process monitor
ps -eaf	Processes	All running processes
kill -9 PID	Processes	Force kill process
nohup cmd &	Background	Run in background
tail -f log	Logs	Live log monitoring
find . -name '*.txt'	Search	Find files by name
find . -empty	Search	Find empty files
grep 'word' file	Search	Search text in file
grep -v 'word' file	Search	Invert search
grep -r 'word' dir/	Search	Recursive search
sed 's/old/new/g' file	Text	Replace text
sed -i 's/old/new/g' file	Text	Replace in file
sed '/word/d' file	Text	Delete matching lines
awk '{print \$1}' file	Text	Print column
awk -F: '{print \$1}' file	Text	Custom delimiter
cut -d: -f1 file	Text	Extract column
tr [a-z] [A-Z] < file	Text	Convert case
sort file	Text	Sort lines
uniq file	Text	Remove duplicates
diff file1 file2	Text	Compare files
ping google.com	Network	Test connectivity
curl https://url	Network	HTTP request
wget URL	Network	Download file
ssh -i key.pem user@host	Network	Remote login
tar -czvf arc.tar.gz dir/	Archive	Create archive
tar -xvf archive.tar.gz	Archive	Extract archive
sudo apt update	Packages	Update package list

Command	Category	What it does
sudo apt install pkg -y	Packages	Install package
systemctl status nginx	Services	Check service status
sudo systemctl start nginx	Services	Start service
chmod 755 file	Permissions	Set file permissions
chmod -R 777 folder/	Permissions	Recursive permissions
chown user:group file	Ownership	Change owner
sudo useradd username	Users	Create user
sudo passwd username	Users	Set password
sudo userdel username	Users	Delete user
sudo groupadd groupname	Groups	Create group
sudo usermod -a -G grp usr	Groups	Add user to group
crontab -e	Scheduling	Edit cron jobs
history	History	Show past commands
time command	Performance	Measure execution time